CSC426 WEEKLY DEV · WEEKEND 3

IRO — Issues Reporting Outlet

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> **Document:** Compact project submission

> **Version:** 1.1 · 19 October 2026

> **Audience:** AAUA students and nearby residents

> **Communities:** Akungba, Iwaro, Etioro, Ayegunle

Live prototype: https://scienceshephard.github.io/CSC426-IRO-220404066/app/

Public repository: github.com/scienceshephard/CSC426-IRO-220404066

1. Purpose and scope

IRO provides a proposed, structured route for students and local residents to report non-emergency campus and community issues, receive a reference, and follow safe status updates. A coordinator would review, categorize, route, and update reports. Public information must not expose a reporter or sensitive details.

The submitted interface is a small front-end prototype. It demonstrates issue entry, validation, anonymity by default, a local reference, and a filterable issue board. It does not transmit reports, authenticate users, or provide a monitored complaint or emergency service.

2. Component interaction

Reporter selects a group, category, broad area, and urgency, then describes the issue → IRO validates and returns a reference → this prototype saves only in the browser → a future secure server routes the case to a coordinator → the coordinator publishes a safe status update.

The initial issue cards and sample survey responses are fictional. Do not enter real or urgent incident information into the prototype.

3. System requirements

Users: reporters (AAUA students and residents/indigenes), IRO coordinators, and system administrators. Coordinators and administrators require authenticated, role-checked access in a production deployment.

3.1 Functional requirements

ID	Requirement	Scope
FR-01	Select reporter group, issue category, and broad area, including AAUA, Akungba, Iwaro, Etioro, and Ayegunle.	Prototype
FR-02	Require a short title, description, and urgency; validate required fields and enforce length limits.	Prototype
FR-03	Make anonymous reporting the default; display privacy guidance and require acknowledgment before saving.	Prototype
FR-04	Save a practice report in local browser storage and show a reference; clearly disclose that it was not sent.	Prototype
FR-05	Display sample/local reports without contact details and filter all, in-progress, and resolved items.	Prototype
FR-06	Authenticate a coordinator; review scope, duplicates, sensitive details, and risk; categorize and assign a case.	Proposed
FR-07	Record timestamped status changes and publish approved, non-identifying updates linked to a reference.	Proposed
FR-08	Show authorized staff aggregate counts and suppress small or sensitive groups that could identify a person.	Proposed

ID	Requirement	Scope
FR-09	Provide an anonymous needs-assessment questionnaire; analyze only consented real responses as community feedback.	Survey

“Prototype” requirements are demonstrated by the submitted front end. “Proposed” requirements need a secure server, database, coordinator workflow, and operational owner.

4. Quality, data, and limitations

4.1 Non-functional requirements

Area	Requirement
Privacy	Collect the minimum information needed; anonymous by default; keep private notes and contact details out of public views.
Security	Production case details require authenticated role access, HTTPS, protected storage/backups, and audit records. The local demo is not secure storage.
Safety	State that IRO is not an emergency service. Add only a verified urgent contact route before production.
Usability/ accessibility	Plain language, visible labels, keyboard access, readable contrast, assistive-technology status feedback, and reduced-motion support.
Responsive design	Support common mobile and desktop widths without horizontal page scrolling.
Performance	Proposed public-page render target: under 3 seconds at the 95th percentile on broadband; measure after hosting.
Integrity/ maintenance	Production status changes are timestamped; categories and routing rules are controlled and documented.
Availability	Production support hours and uptime target must be agreed with the responsible university/community organization.

4.2 Core data objects

IssueReport: reference, group, category, broad area, title, description, urgency, status, created time, anonymity. **StatusUpdate:** report reference, status, public message, time, coordinator. **Coordinator:** account and role. **SurveyResponse:** pseudonymous response ID, group, broad area, and answers. Keep optional contact information separate and restricted in any production system.

4.3 Constraints

The demo uses HTML, CSS, JavaScript, and browser local storage. It has no server, authentication, notifications, file uploads, moderation, case assignment, or real status service. Browser storage may be visible to other users of the same browser profile. Production needs approved routing owners, privacy/retention rules, access control, secure storage, and a staffed response process.

QUESTIONNAIRE AND PROBLEM ANALYSIS

5. Community needs assessment

Audience: AAUA students and residents/indigenes of Akungba, Iwaro, Etioro, Ayegunle, and nearby areas. The anonymous questionnaire covers group and broad area, recent issues, issue types/frequency, reporting channels/barriers, privacy confidence, desired updates, response expectations, and optional non-identifying suggestions. It requests no name, matric number, phone, exact address, or email.

> **Form status:** The questionnaire and Google Apps Script are in the repository. A live Google Form responder URL must be generated by running `createIROSurvey` from a signed-in Google account before distributing the survey.

5.1 Simulated sample analysis

The CSV contains 15 fictional rows for coursework demonstration only; no community responses are claimed. Group mix: 9 students, 3 Akungba residents, and one each from Iwaro, Etioro, and Ayegunle. Issue types were multi-select, denominator 15.

Issue type	Count	Issue type	Count
Sanitation	10 (66.7%)	Roads	8 (53.3%)
Safety	7 (46.7%)	Utilities	6

Issue type	Count	Issue type	Count
			(40.0%)
Facilities	5 (33.3%)	Communication	4 (26.7%)
Transport	4 (26.7%)	Top barrier: do not know contact	8 (53.3%)

Other simulated barriers: expect no follow-up 6/15; privacy concern 6/15; process takes time 4/15. Multi-select percentages may total over 100%. These fictional counts are not representative findings.

5.2 Ishikawa summary

Effect investigated: fragmented, delayed, and unsafe reporting, with low trust and weak follow-up. Possible causes are people (awareness/ownership/fear), process (separate channels/no references), technology (accessibility/connectivity), governance and trust (unclear privacy/response owner), communication (no updates), and resources/environment (staff, funds, distance, infrastructure). Validate these hypotheses with genuine feedback.

USE-CASE SPECIFICATION

6. Actors and main use cases

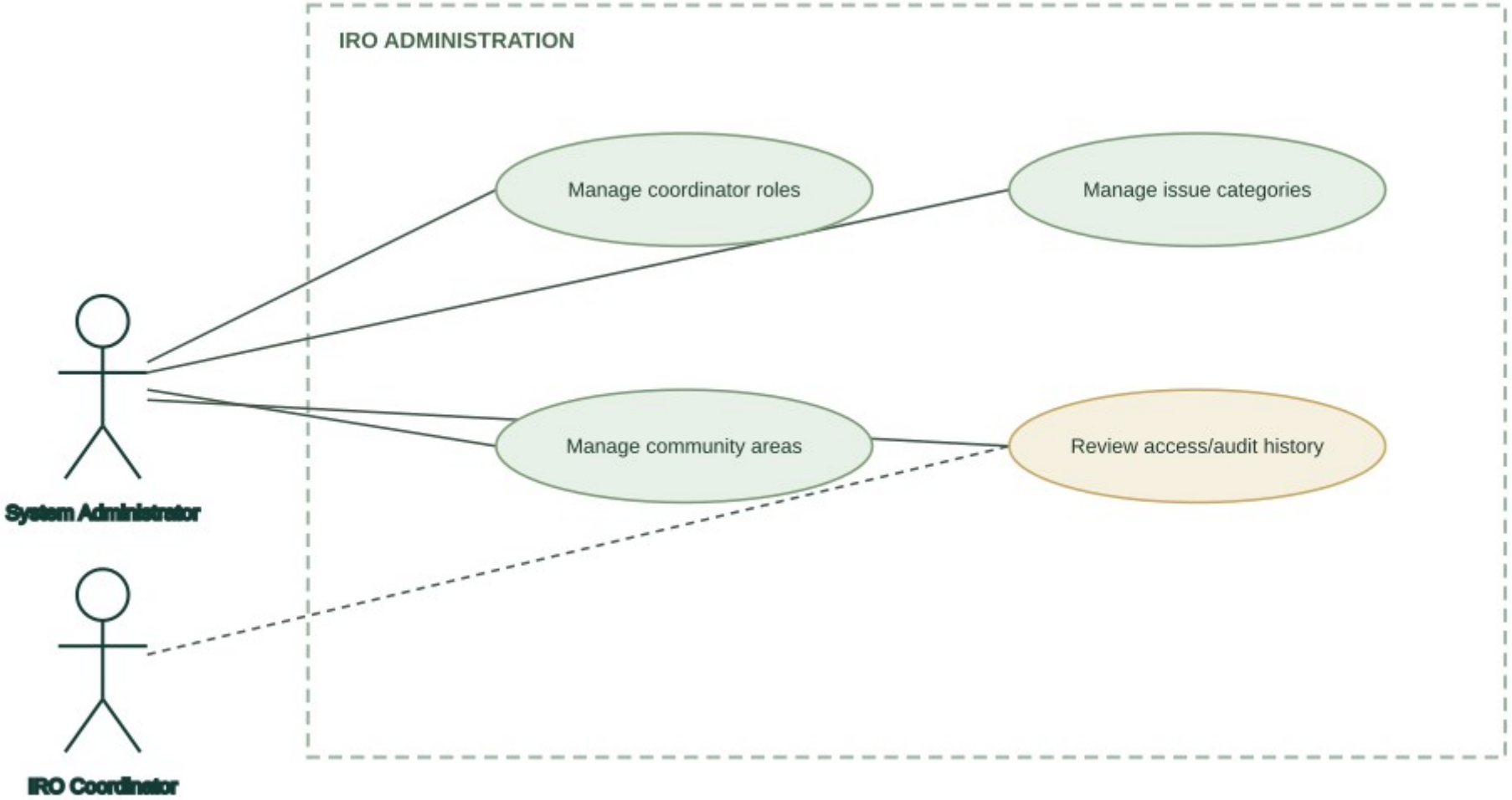
Actors: Reporter (AAUA student or resident), IRO Coordinator, and System Administrator. Figures 1–2 show reporting and status tracking, coordinator review/routing/updates, aggregate summaries, and administrative access/category management. The detailed use-case narratives and editable draw.io files are in the repository.

USE CASE · REPORTING AND CASE MANAGEMENT



Proposed operational system; coursework prototype implements local report entry and a sample status board only.

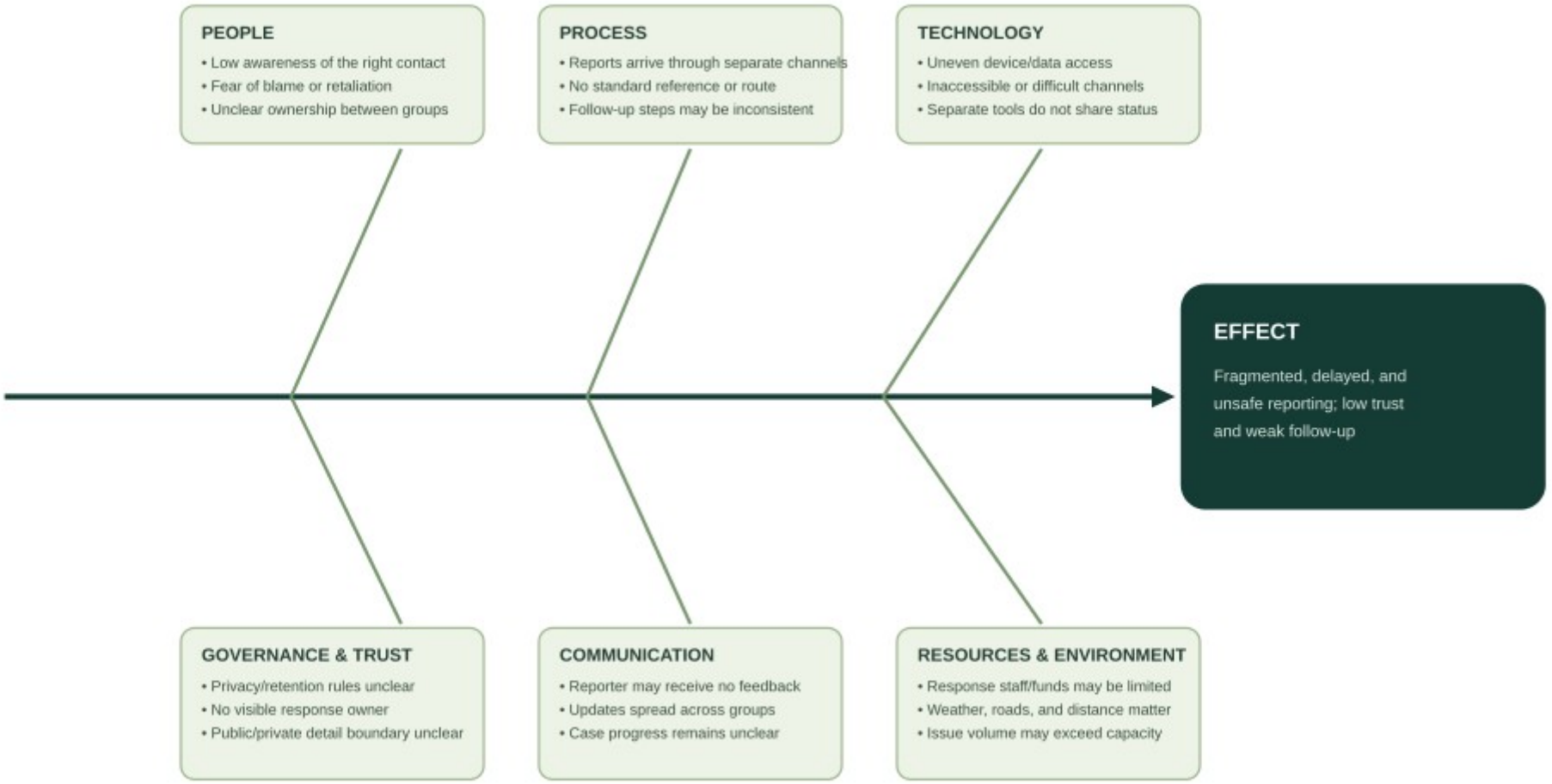
USE CASE · ADMINISTRATION



Production administration requires authenticated and role-checked access.
The coursework prototype does not implement administrative functions.

IRO · ISHIKAWA (FISHBONE) PROBLEM ANALYSIS

Potential causes to validate with genuine community feedback



Potential causes are hypotheses for validation, not measured community findings.

OBJECT LISTING AND SOURCE REFERENCES

7. Objects and code

ReportForm
Captures group, category, broad area, title, description, urgency, and privacy acknowledgment.

LocalReportStore
Stores practice reports in browser `localStorage`; not a secure shared database.

StatusUpdate
Proposed status, public message, time, and coordinator reference.

SurveyResponse
Pseudonymous survey record; included sample rows are simulated.

IssueReport
Reference, category, broad area, title, description, status, date, urgency, anonymity/sample flags.

IssueBoard
Renders safe report fields, counts, and status filters.

Coordinator / Category
Proposed authenticated staff role and controlled issue category.

IRO API / Database
Proposed server validation, protected storage, access checks, references, and audit history.

Code and project links

- **Live website:** scienceshephard.github.io/CSC426-IRO-220404066/app/
- **Full repository:** github.com/scienceshephard/CSC426-IRO-220404066
- **Web source code:** app/ (HTML, CSS, JavaScript)
- **Google Forms creation script:** CreateIROForm.gs
- **Editable diagrams and full SRS:** diagrams/ · SRS.md

> **Printing note:** This compact document embeds all three diagrams. The long source listing is not printed; the code is available at the links above. Print this document at 10 pt. Survey results in the CSV are simulated, not community feedback.